Simple Test for Dexterous Hand Initialization Failure

Controlled Status:

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Department						
Name						

Release Date: Effective Date:

1 Purpose

To verify whether a power short circuit is the cause when the dexterous hand shows no movement after power-on.

2 Operating Procedure

2.1 Test the Power Board for Short Circuits

1. As shown in Figure 1，test at the TVS diode (input power) on the power board.

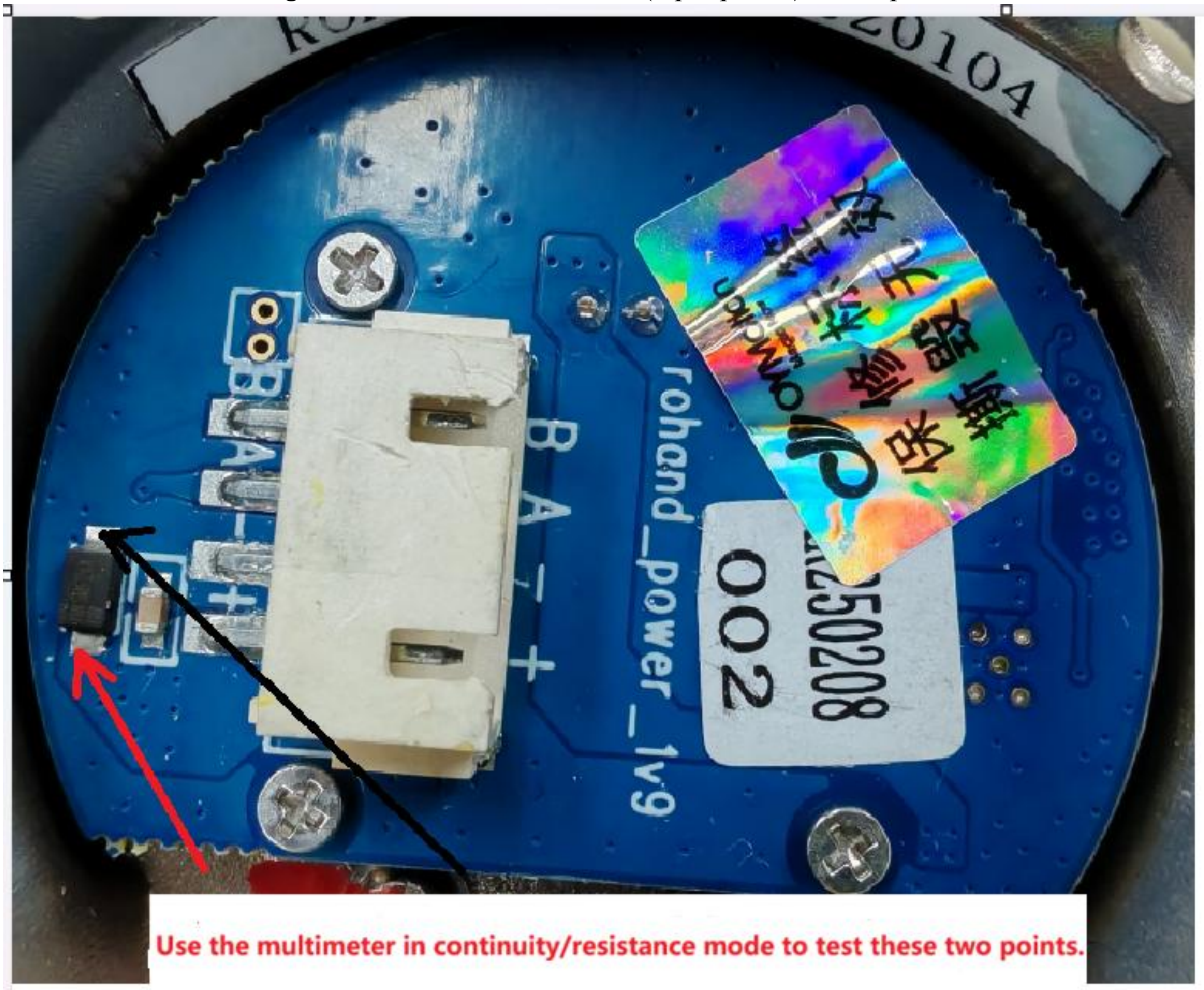


Figure 1

2. Set the multimeter to resistance/continuity mode. Test as shown in Figure 2. The reading should be open (infinite resistance, not near 0 Ω).:

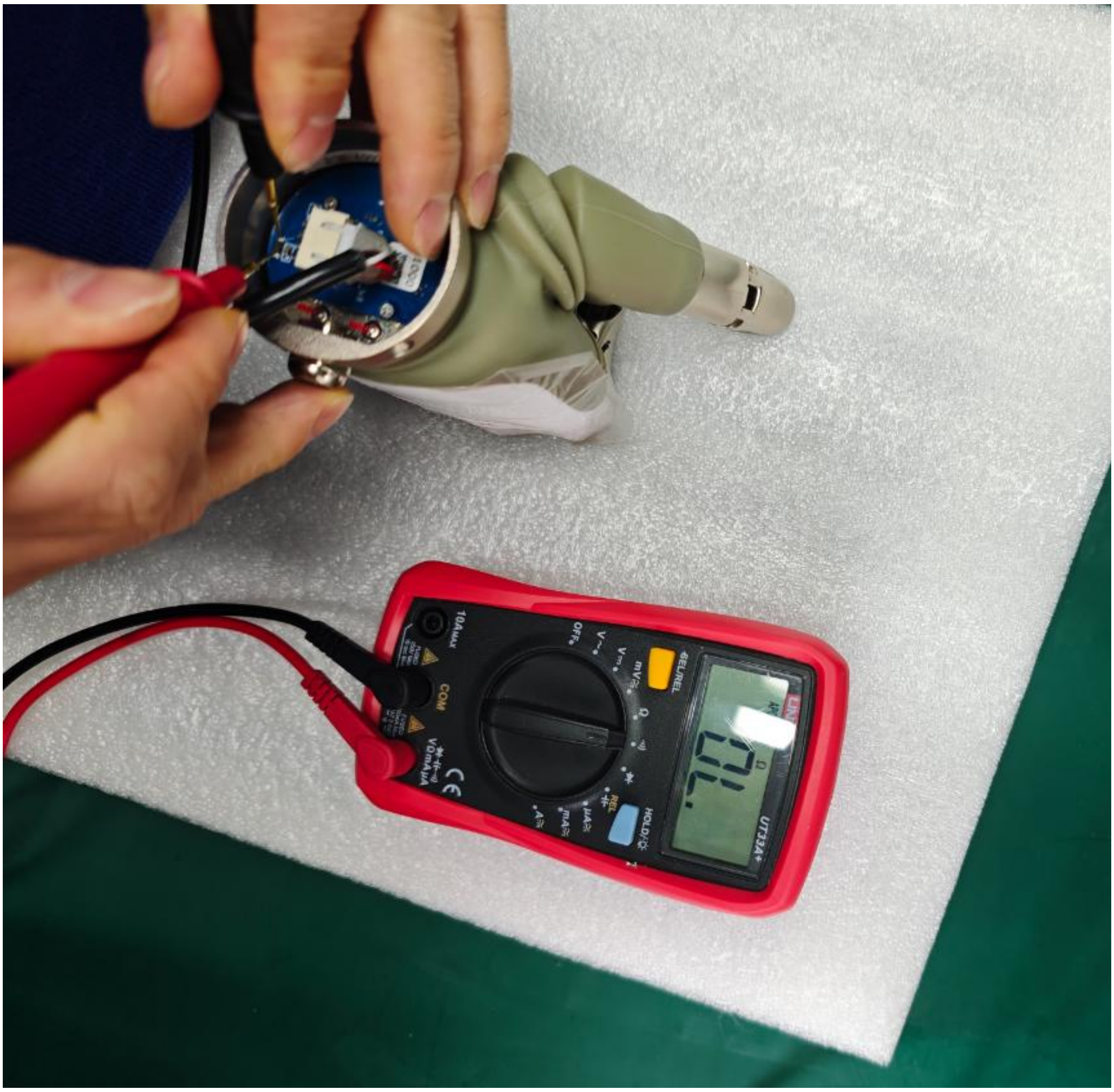


Figure 2

3. Voltage measurement points from the power board to the hand board, as shown in Figure 3:

Figure 3

4. Set the multimeter to resistance/continuity mode. Test as shown in Figure 4. The reading should be open (infinite resistance, not near 0 Ω).:

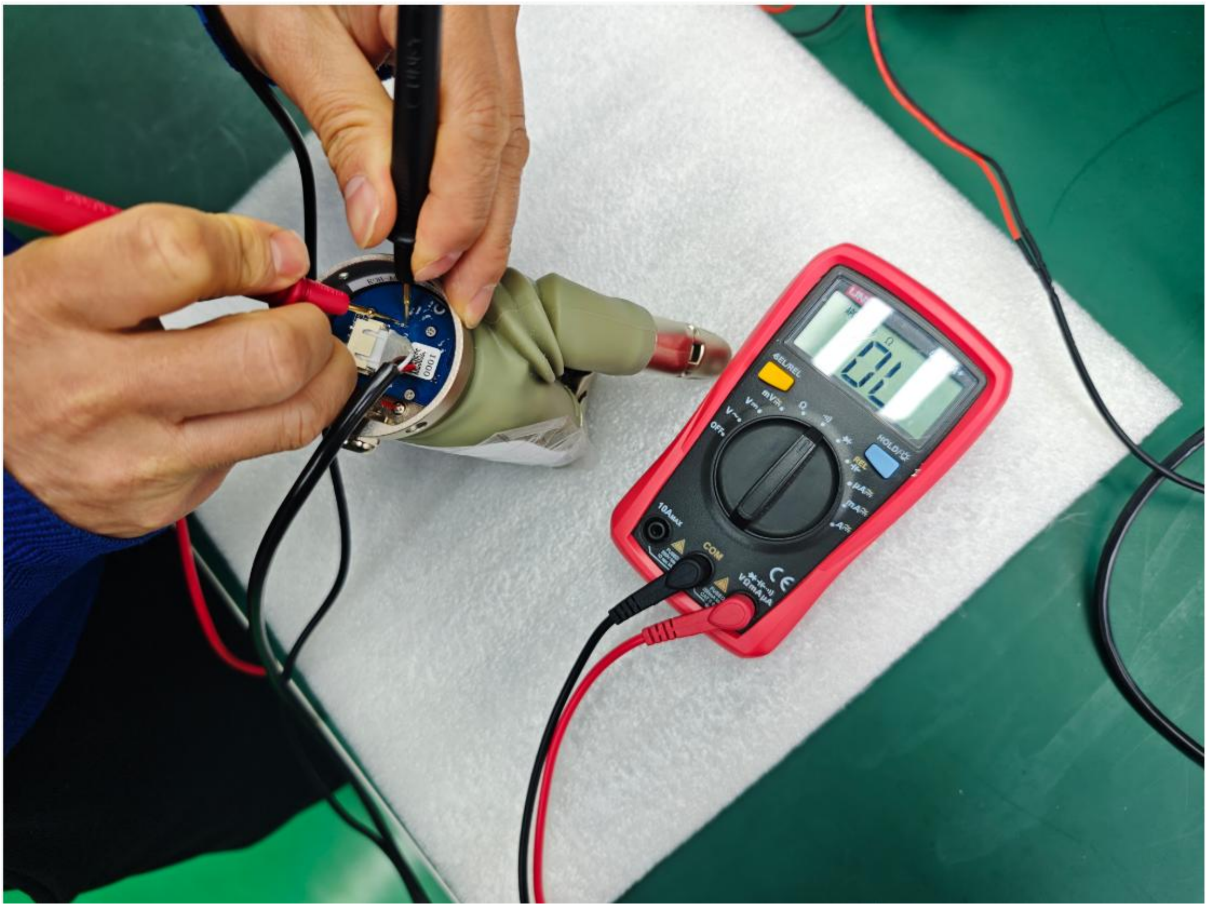
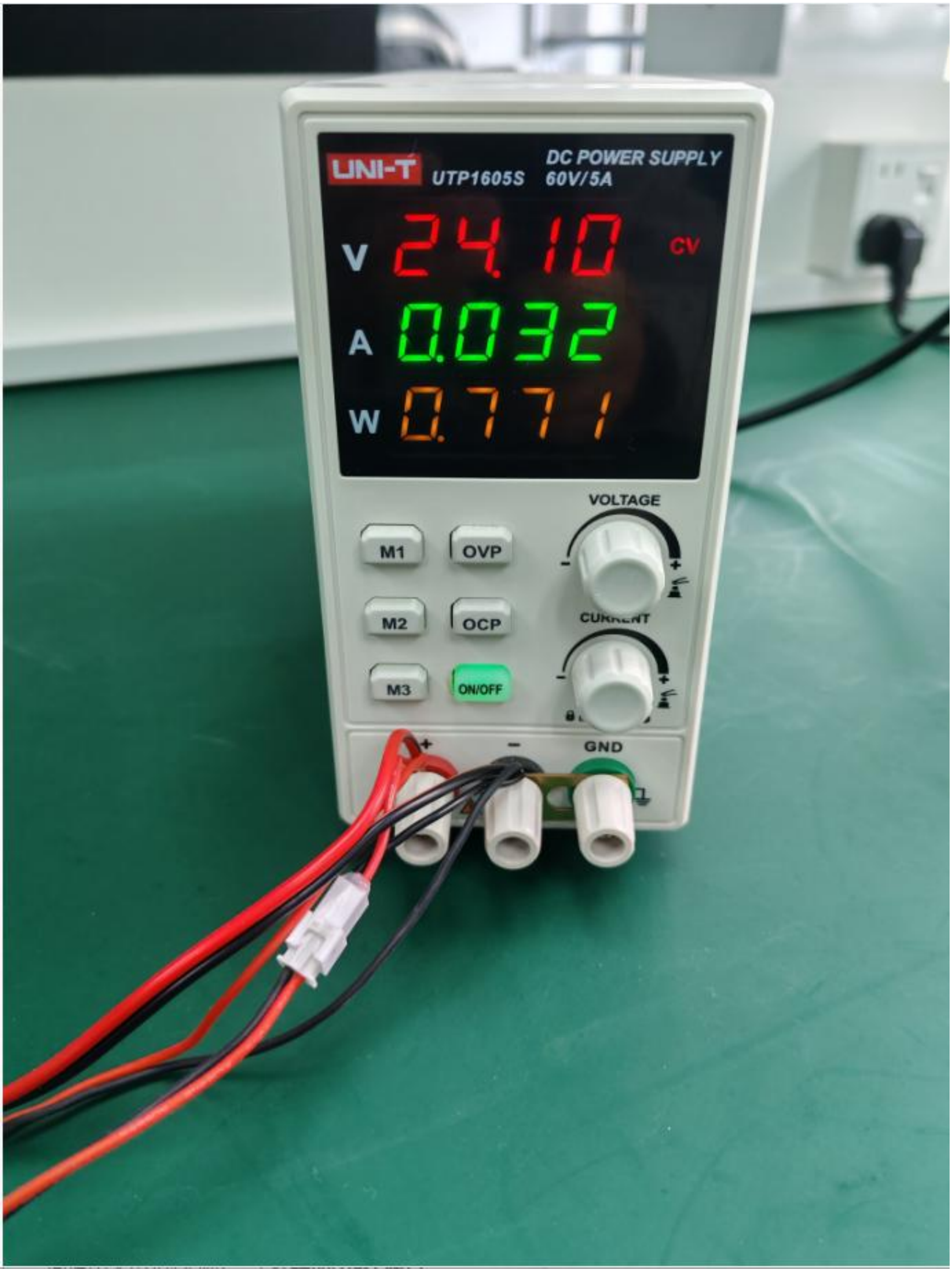


Figure 4

2.2 If no short circuit is found on the power board, proceed with power-on testing.:

1. If a regulated power supply is available, connect it as shown in Figure 5. Apply 24 V. The current after power-on should be approximately 32 mA.



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2. Test the power input on the power board. The reading should be approximately 24 V, as shown in Figure 6.

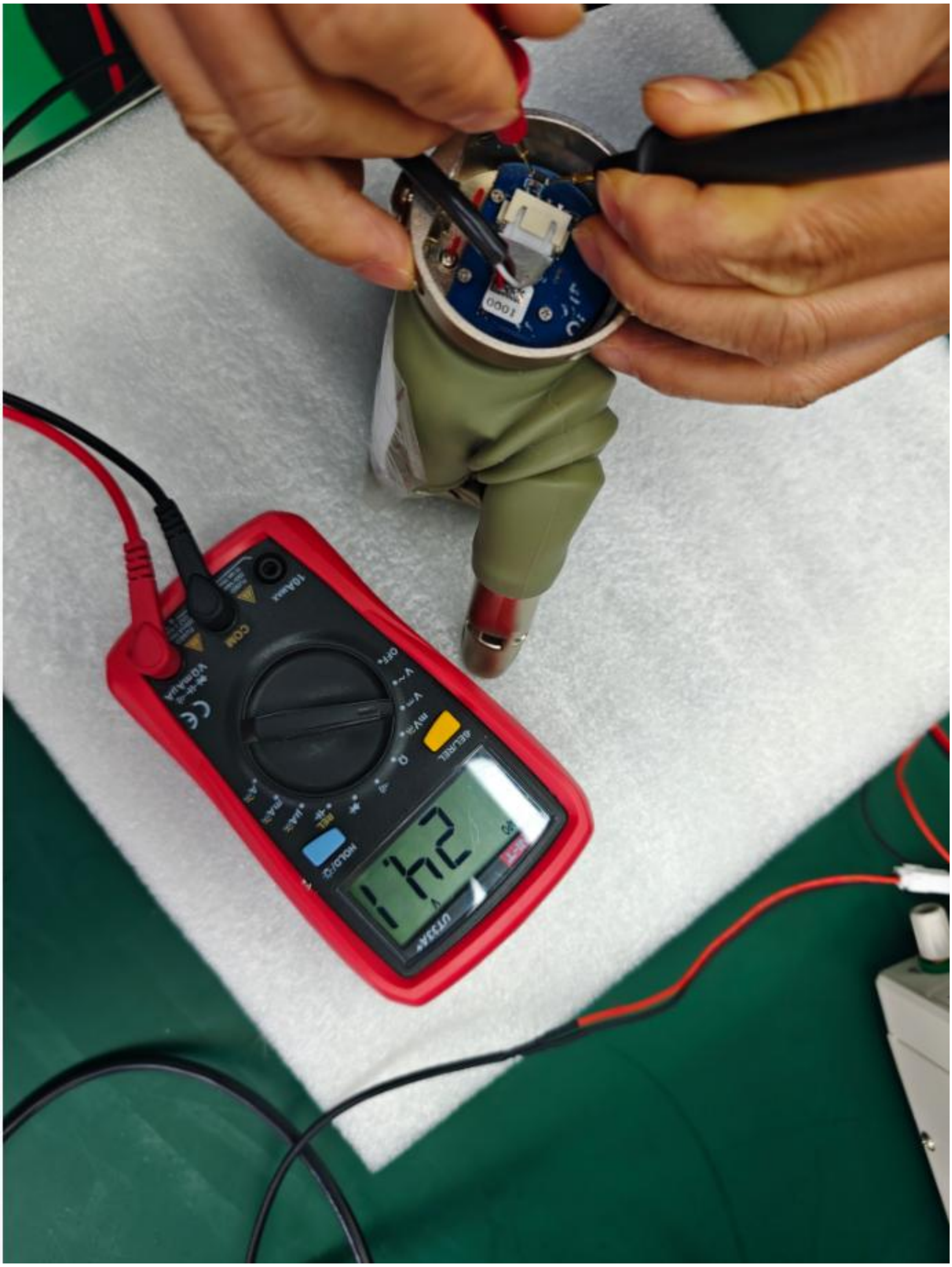

Figure 6

3. Test at the power output to the hand main board on the power board. The voltage should be approximately 11.2 V (10.5 V to 12 V is acceptable), as shown in Figure 7.:

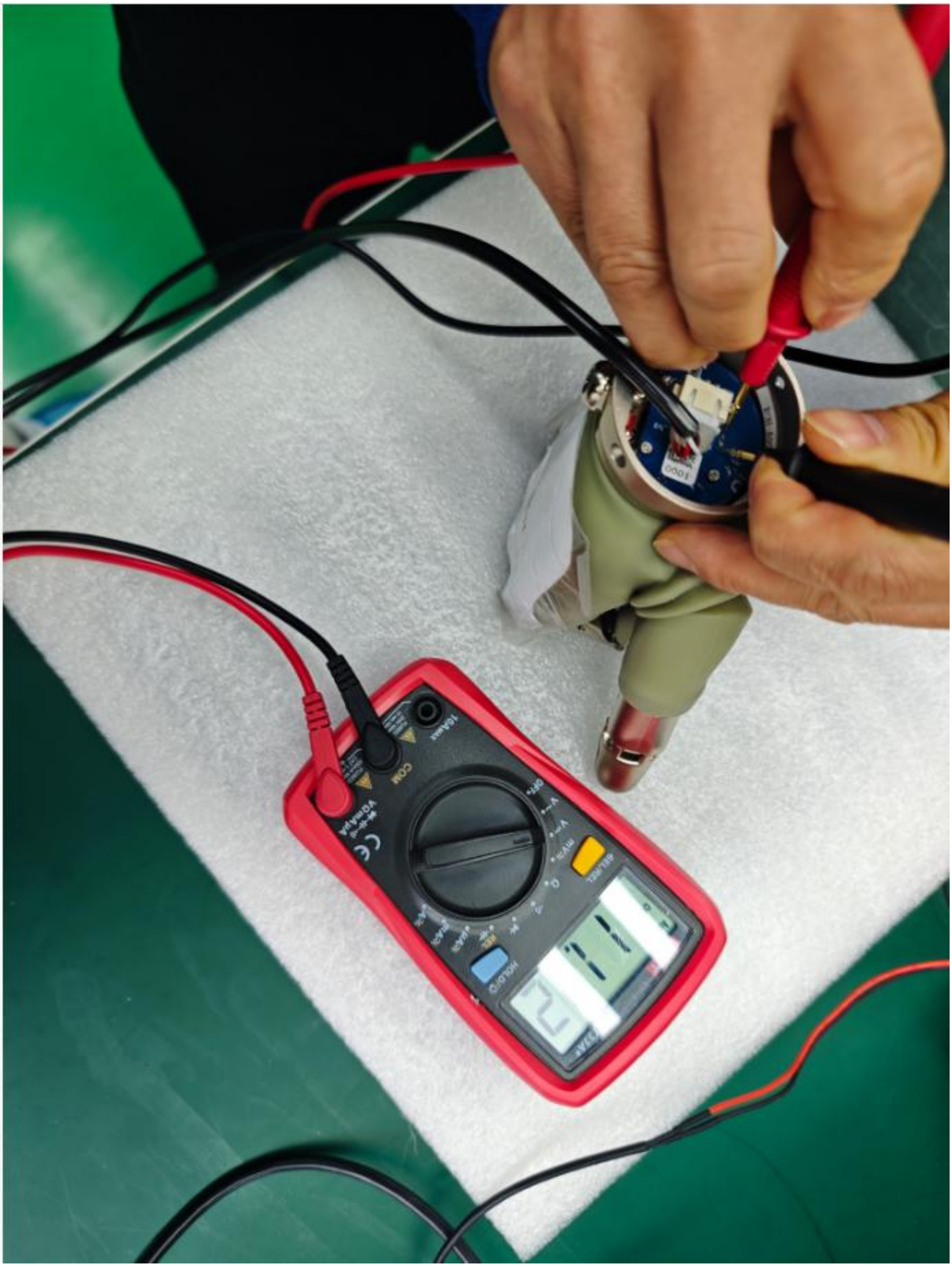

Figure 7

4. If all of the above are correct, return to the manufacturer for analysis.